Build a measured driving lane

Name: ____________________ Team: ____________________ Date: __________

A short, clear test lane for measuring the robot’s travel.

Gather these materials

[ ] Checked wheeled base and connected device

[ ] Ruler or measuring tape

[ ] Removable floor tape

[ ] Paper and pencil

Build: steps 1-4

[ ] 1. Choose a flat floor space with the teacher. Keep it away from stairs and table edges. Make the lane at least twice as wide as the robot.

[ ] 2. Mark a start line across the lane. Put the ruler beside the lane, with zero at the start line. Keep the ruler out of the wheel path.

[ ] 3. Add a small tape mark on the side of the robot, above its main wheel axle. Use this same mark to measure every run. An axle is the rod a wheel turns on.

[ ] 4. With the power off, turn each wheel gently by hand. Remove anything rubbing a wheel. Check that the sensor mount stays clear of the floor.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

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Build: steps 5-8

[ ] 5. Raise the wheels on firm supports. Try the short program below. If the wheels would send the robot backward, stop and ask the teacher to correct the motor direction.

[ ] 6. Place the robot's measuring mark over the start line. Clear the lane. With the teacher's approval, run the program and wait until the wheels stop.

[ ] 7. Measure from the start line to the same mark on the stopped robot. Write the distance in centimeters (cm). Return it to the start with the power off.

[ ] 8. Repeat three times with the same code. Then ask the teacher to help test the Stop control during another short run. Save the lane and measuring mark.

Make the program

Start with movement stopped. Set low speed, such as 20 percent.

Move forward for 0.8 seconds, then stop.

End the program. Do not use a forever-repeat block.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Speed setting	Distance (cm)	Stopped by itself?
1			
2			
3			

A successful build

The robot stops by itself after each short drive.

All three distances are recorded from the same mark.

The teacher checks the screen Stop control and power-off method.

Why might three runs travel slightly different distances?